GRADE 5

STEM OLYMPIAD

National Level Examination (NLE)

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

Sample Paper

SECTION A NUMBER SENSE AND OPERATIONS Q1 TO Q17

1. Observe the pattern: 5, 11, 23, 47, 95, ____. Which number should replace the blank?
A. 181
B. 191
C. 192
D. 193

2. A number is 7,00,000. If 1,999 is added to its predecessor, what number is obtained?
A. 7,01,998
B. 7,01,999
C. 7,02,000
D. 6,99,999

3. Which Roman numeral represents the difference between 94 and 47?
A. XLVII
B. XLIX
C. LIV
D. XLIV

4. If 3 days before a certain day was Thursday, what day will it be 11 days after that day?
A. Monday
B. Tuesday
C. Wednesday
D. Thursday

5. A quadrilateral has exactly one pair of parallel sides and its two non-parallel sides are equal. Which description fits it?
A. Square
B. Rectangle
C. Isosceles trapezium
D. Rhombus

6. Which is the greatest number?
A. 7,08,999
B. 7,89,090
C. 7,80,999
D. 7,89,009

7. A school needs exactly 2,50,000 pencils. It already has 1,76,850 pencils. How many more are needed?
A. 72,150
B. 73,150
C. 74,150
D. 73,050

8. Find the missing number: 2, 6, 18, 54, 162, ____. Which operation describes the pattern?
A. 486; multiply by 3
B. 324; multiply by 2
C. 486; add 324
D. 648; multiply by 4

10. What is the smallest six-digit number that is divisible by 9?
A. 1,00,008
B. 1,00,009
C. 1,00,017
D. 1,00,018

11. A parcel weighs 2 kg 250 g. Another weighs 1 kg 875 g. What is their total weight?
A. 4 kg 25 g
B. 4 kg 125 g
C. 4 kg 75 g
D. 3 kg 1,125 g

12. A boy faces South-East. He turns 90° clockwise and then 180° anticlockwise. Which direction does he finally face?
A. North-East
B. North-West
C. South-West
D. South-East

13. Four children stand in a line. Rohan is taller than Karan. Karan is taller than Mohit. Aryan is taller than Rohan. Who is second tallest?
A. Mohit
B. Aryan
C. Rohan
D. Karan

14. A number lies exactly halfway between 3,475 and 4,125. What is the number?
A. 3,750
B. 3,800
C. 3,825
D. 3,850

15. The smallest three-digit number is multiplied by the largest one-digit prime number. What is the result?
A. 700
B. 693
C. 707
D. 900

16. A movie lasts 2 hours 45 minutes. If it starts at 10:35 a.m., when does it end?
A. 1:10 p.m.
B. 1:20 p.m.
C. 1:25 p.m.
D. 12:20 p.m.

17. Which number is divisible by both 2 and 5, but not by 10?
A. 40
B. 55
C. 70
D. No such whole number

Sample Paper

9. A number 58,42,716 is increased by 2,00,000. What is the place value of the digit 4 in the new number?
A. 4,000
B. 40,000
C. 4,00,000
D. 40,00,000

SECTION B MEASUREMENT GEOMETRY AND REASONING Q18 TO Q34

18. Which number is formed by 8 lakh + 6 ten-thousands + 4 thousands + 9 tens + 7 ones?
A. 8,64,097
B. 8,64,970
C. 8,46,097
D. 8,60,497

19. If 96 ÷ 8 = 12, which expression has the same value as 96 ÷ 8 × 7?
A. 12 × 7
B. 96 ÷ 56
C. 8 × 7
D. 96 ÷ 15

20. A solid has 6 equal square faces, 12 edges and 8 vertices. If each edge is 4 cm, what is the total length of all its edges?
A. 24 cm
B. 36 cm
C. 48 cm
D. 64 cm

21. A is the sister of B. B is the father of C. D is the mother of A. How is A related to C?
A. Mother
B. Aunt
C. Sister
D. Grandmother

22. Three of the following expressions have the same value. Which one is different?
A. 9 × 6
B. 72 − 18
C. 18 + 36
D. 96 ÷ 2

23. A clock shows 7:40. What is the smaller angle between its hour hand and minute hand?
A. 10°
B. 20°
C. 30°
D. 40°

24. Find the next number: 250, 235, 205, 160, 100, ____.
A. 35
B. 25
C. 30
D. 40

27. Complete the letter pattern: C, F, J, O, U, ____.
A. A
B. B
C. C
D. D

28. On a map, 1 cm represents 8 km. Town P and Q are 4.5 cm apart on the map. If a traveller goes from P to Q and returns by the same route, what total distance does he travel?
A. 36 km
B. 64 km
C. 72 km
D. 80 km

29. Which fraction is greatest?
A. 7/10
B. 5/7
C. 9/13
D. 11/16

30. A number is 25 more than 6 times 125. What is the number?
A. 750
B. 775
C. 800
D. 825

31. A 4 × 4 square tile wall is numbered from 1 to 16 row-wise. All tiles containing odd numbers are removed. How many tiles remain, and what fraction of the wall remains?
A. 8 tiles, 1/2
B. 7 tiles, 7/16
C. 8 tiles, 1/4
D. 9 tiles, 9/16

32. The distance between two towns is more than 15 km but less than 20 km. It is also more than 18 km but less than 19.5 km. If the distance is a whole number, what is it?
A. 17 km
B. 18 km
C. 19 km
D. 20 km

33. A sheet is folded in half repeatedly. Each fold doubles the number of equal layers. How many folds are needed to make 32 equal parts?
A. 4
B. 5
C. 6
D. 8

Sample Paper

25. A girl is facing North. She turns right, then makes a half-turn, and then turns left. Which direction is she facing?
A. North
B. South
C. East
D. West

26. Which sequence does NOT follow the same rule as the other three?
A. 4, 9, 14, 19, 24
B. 6, 12, 18, 24, 30
C. 7, 14, 21, 28, 35
D. 3, 8, 13, 19, 23

SECTION C APPLIED MATHEMATICS Q35 TO Q45

35. A shop is 2 km 350 m from a school. A library is 1 km 750 m farther from the school than the shop. If a child walks school → shop → library, what distance does the child cover?
A. 5 km 450 m
B. 6 km 450 m
C. 6 km 850 m
. 5 km 850 m

36. The playground is 600 m east of the school. The library is 400 m north of the playground. The library is therefore in which direction from the school?
A. North-East
B. North-West
C. South-East D. South-West

37. Rahul walks 6 steps North, 8 steps East, 6 steps South and 3 steps West. How far and in which direction is he from his starting point?
A. 5 steps East
B. 5 steps West
C. 3 steps East
D. 3 steps West

38. A teacher has 1,250 stickers and wants to give the same number to 6 children, keeping the remainder for herself. What is the greatest number each child can receive?
A. 206
B. 207
C. 208
D. 209

39. A number divided by 8 gives quotient 125 and remainder 7. What is the number?
A. 997
B. 1,000
C. 1,007
D. 1,017

40. 1,152 chocolates are packed equally into boxes. Each box holds 24 chocolates. If 3 boxes are kept aside for an event, how many boxes are left for regular distribution?
A. 45
B. 46
C. 47
D. 48

43. A train journey lasts 3 hours 45 minutes. If the train reaches at 7:20 p.m., at what time did it leave?
A. 3:25 p.m.
B. 3:35 p.m.
C. 3:45 p.m.
D. 4:35 p.m.

44. Riya studied from 4:35 p.m. to 5:50 p.m. She claims that she studied for 1 hour 25 minutes. Which statement is correct?
A. Yes, exactly 1 h 25 min
B. No, it was 1 h 10 min
C. No, it was 1 h 15 min
D. No, it was 1 h 20 min

45. Three activities occur one after another: Drawing—25 min, Reading—40 min, Games—35 min. If they start at 3:15 p.m. and there is a 10-minute break between each activity, when will everything finish?
A. 4:45 p.m.
B. 4:55 p.m.
C. 5:00 p.m.
D. 5:10 p.m.

SECTION D EXCELLENCE ZONE Q46 TO Q50

46. Complete the sequence: A10, C20, F30, J40, O50, ____.
A. T60
B. U60
C. V60
D. W60

47. Which number does NOT follow the pattern: 4, 9, 16, 25, 36, 48, 64?
A. 25
B. 36
C. 48
D. 64

48. Prateek wants to distribute all his pencils equally between himself and his sister, with no pencil left over. Which number could be the total number of pencils if he has more than 20 but fewer than 30?
A. 21
B. 23
C. 24
D. 27

Sample Paper

41. Find the number that should replace □: 28,750 + □ = 75,000 − 11,250.
A. 34,500
B. 35,000
C. 35,500
D. 36,000

42. A class begins at 9:20 a.m. The teacher teaches for 35 minutes, conducts an activity for 15 minutes, and then teaches for 45 minutes. There is a 10-minute break after the activity. At what time does the class finish?
A. 11:05 a.m.
B. 11:15 a.m.
C. 11:25 a.m.
D. 11:35 a.m.

49. A square is divided into 4 equal smaller squares. In three of them, one diagonal triangle is shaded; in the fourth, two diagonal triangles are shaded. How many shaded triangles are there altogether?
A. 4
B. 5
C. 6
D. 7

50. In a number machine, 654 and A give 777. The same rule is used with 876 and A. What should replace ? if the rule is “add the same number A”?
A. 987
B. 999
C. 990
D. 998

Sample Paper